

CLASS 05 / SEP 22 / U03

FROM THE PYRAMID TO YOUR MODULE

BUILD IT, THEN LIGHT IT — THE P3a SHEET

NOTHING DUE

U03 CONTINUES / P3a IN PROGRESS / DUE WEEK 6 — SEP 29 / 7%

PARAMETERS FIRST

Before a single point: write down the numbers that drive the module, and what each one does.

A PARAMETER TABLE

NAME	VALUE	WHAT IT CONTROLS
size	1.0	outer edge — also the tiling pitch
thickness	0.1	depth of the panel
aperture k	0.5	hole as a share of the size, $0 < k < 1$

ONE MUST CHANGE THE SHAPE

Size and thickness only change the box. P3a asks for a shape control beyond overall dimensions:

APERTURE · FOLD DEPTH · CORNER OFFSET

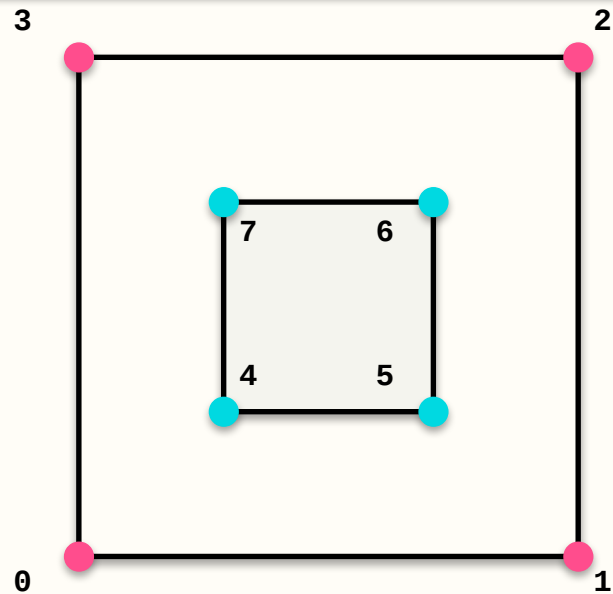
The example here is an aperture: a square frame with a square hole. Yours comes from your precedent.

IF YOU CANNOT SAY WHAT A NUMBER CONTROLS, IT IS NOT A PARAMETER YET.

CALCULATE EVERY POINT

Each position is an expression of the parameters. Never a typed coordinate.

PLAN — TOP RING



THE POINT LIST

```
h = size / 2           # half size
q = h * k              # aperture
t = thickness
```

```
top    = [[-h, -h, t], [ h, -h, t],      # 0 1
          [ h,  h, t], [-h,  h, t],      # 2 3
          [-q, -q, t], [ q, -q, t],      # 4 5
          [ q,  q, t], [-q,  q, t]]      # 6 7
```

```
bottom = the same eight at z = 0        # 8-15
verts  = top + bottom
```

WRITE THE INDEX NEXT TO EVERY POINT. THE FACES ARE LISTS OF THESE NUMBERS.

FACE ORDER AND CLOSURE

Four kinds of face, each written counter-clockwise as seen from outside the solid.

THE FACE LIST — $i = 0 \dots 3$, $n = (i + 1) \% 4$

TOP

[i , n , $4+n$, $4+i$]

normal up

BOTTOM

[$12+i$, $12+n$, $8+n$, $8+i$]

normal down

OUTER

[i , $8+i$, $8+n$, n]

normal outward

INNER

[$4+i$, $4+n$, $12+n$, $12+i$]

normal into the hole

CHECK IT

16 VERTICES · 32 EDGES · 16 FACES

Closed: every edge is shared by exactly two faces.

Oriented: a shared edge runs one way in one face and the other way in the next.

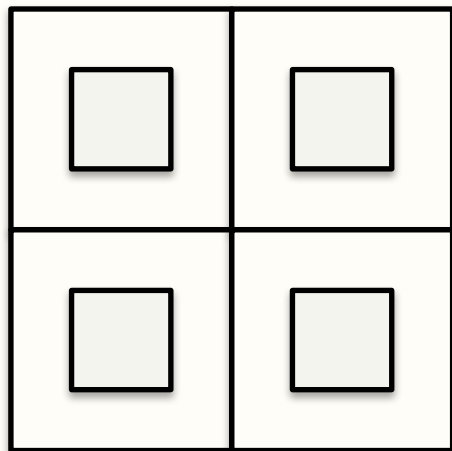
A reversed face flips its normal. Show one on the sheet, and say how you found it.

PUT ONE FACE ON THE SHEET WITH ITS VERTEX ORDER DRAWN. P3a ASKS FOR IT.

TILE IT 2×2 , THEN PUSH THE NUMBERS

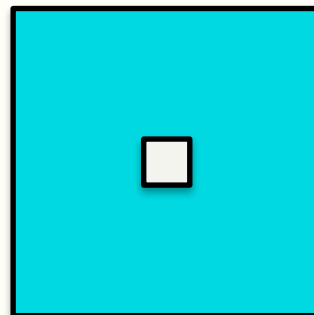
A module that only works at one value is a drawing. Find the range where it still closes and tiles.

2×2 , EDGE TO EDGE

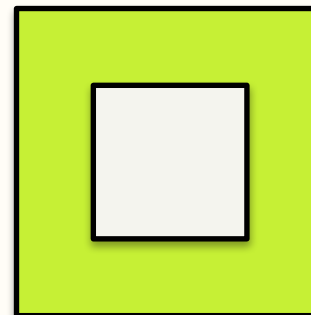


Size is the pitch. Shared edges, no gaps.

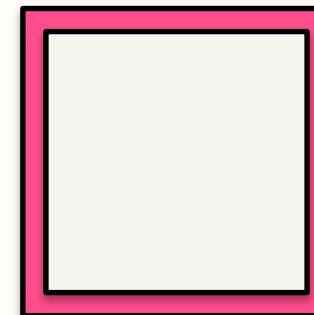
LOW · MIDDLE · HIGH



$k = 0.15$



$k = 0.50$



$k = 0.85$

At $k = 0$ the hole is gone and the inner faces collapse. At $k = 1$ the frame has no width. Both ends break the mesh.

STATE THE RANGE ON THE SHEET — FOR EXAMPLE $0.15 \leq k \leq 0.85$ — AND DRAW BOTH ENDS AND THE MIDDLE.

ONE RENDER, ONE MATERIAL

A studio render: the module alone, on a ground plane, in one surface.

GROUND PLANE

The module sits on it. Its shadow tells the eye where the module is.

Make it large enough to hold the whole shadow.

ONE MATERIAL

One surface for the whole module: colour, roughness, metal or not.

Write the values down. They go on the sheet.

BACKGROUND

Plain and continuous. The floor meets the backdrop without a visible edge.

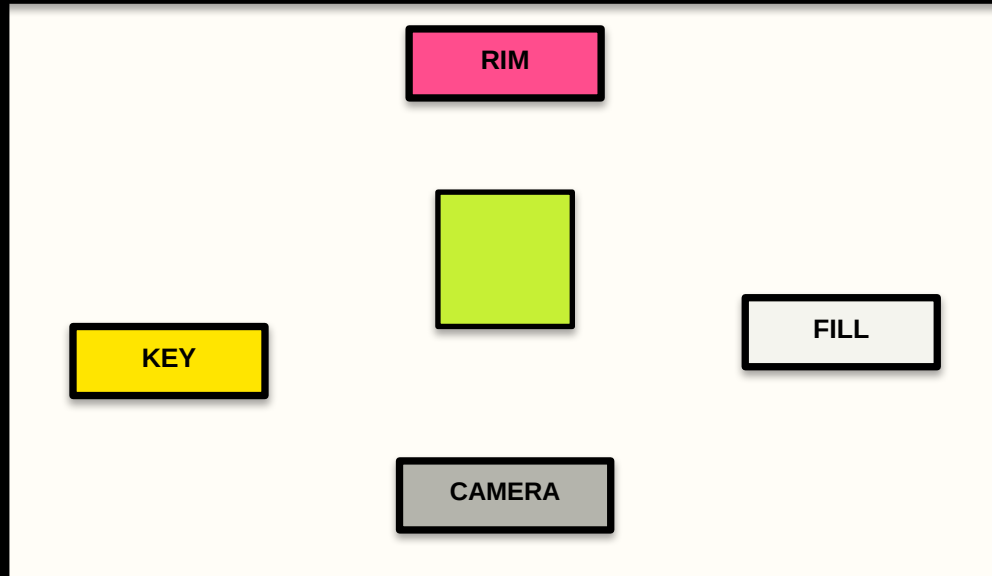
No entourage: no people, trees or context.

THE GEOMETRY IS THE SUBJECT. THE MATERIAL SHOULD SHOW ITS FORM, NOT COMPETE WITH IT.

LIGHT IT ON PURPOSE

Choose one setup. Every light has a job, and you can name it.

KEY · FILL · RIM (PLAN)



TWO SETUPS

KEY the main light. It sets the shadow direction.

FILL opposite side, weaker. It lifts the dark side.

RIM behind the module. It separates the edge from the background.

OR

SUN one direction, sharp shadow.

SKY soft light from everywhere that fills the shadow.

LARGE LIGHT, SOFT SHADOW. SMALL LIGHT, SHARP SHADOW. CHOOSE ONE, SAY WHY, AND RECORD EVERY SETTING.

CAMERA AND SHEET

One fixed view, then a layout where the render leads and the lists explain it.

CAMERA

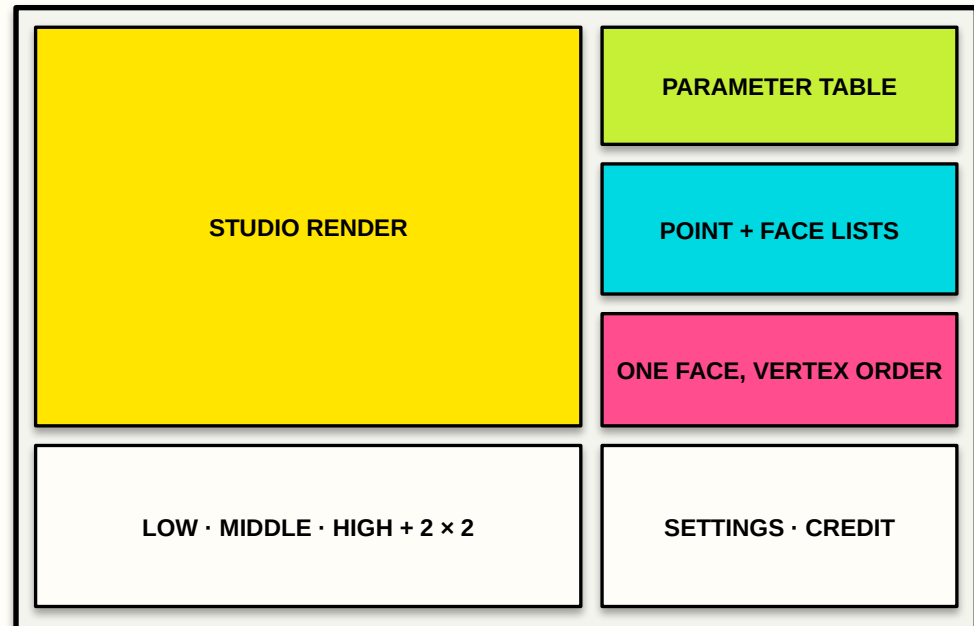
One view. Slightly above, three-quarter, so the top and two sides read.

A long enough lens that the module does not distort.

Frame the module and its shadow. Leave air around both.

Render at the proportion of the space it fills on the sheet.

17 × 11 INCH SHEET



SETTINGS = MATERIAL VALUES AND EVERY LIGHT. CREDIT = BUILDING, ARCHITECT, SOURCE.

CRITIQUE CHECKLIST

What we look at when your sheet goes on the wall. Check it yourself first.

THE MODULE

- ☐ Tiles 2×2 , edge to edge, with no gaps
- ☐ Closed: every edge shared by two faces
- ☐ One shape parameter beyond the box
- ☐ Ranges stated; low, middle and high drawn
- ☐ Every point calculated, none typed

THE RENDER

- ☐ Sits on the ground; the shadow reads
- ☐ One material, its values recorded
- ☐ Each light has a job you can name
- ☐ Plain background, no entourage
- ☐ The render leads the sheet; lists explain it

IF YOU CANNOT TICK A BOX, THAT IS WHAT YOU WORK ON THIS WEEK.

REMINDER — P3a REQUIREMENTS

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Choose a simple panel from a built precedent that tiles edge to edge in a 2×2 grid. Credit the building, architect and source.

02

Define parameters first and explain each. Include a shape control such as aperture, fold depth or corner offset, beyond overall dimensions.

03

Build an editable, closed mesh from explicit point and face lists. Calculate point positions from the parameters.

04

State the ranges where the module closes and tiles. Diagram low, middle and high values of a shape parameter.

05

Use one 17×11 inch sheet with one studio render of the module on a ground plane — one material, a deliberate lighting setup (key, fill and rim, or one sun and sky) and no entourage. Record the material and light settings on the sheet.

06

Beside the render, show point and face lists, a parameter table and the vertex order of one face. Explain normals, closure and reversed face winding.

NOW — BUILD IT, THEN LIGHT IT

Rest of the class: your own module, closed and tiling, in a first test render.

✓

Parameters written down first, with one shape control beyond the box

✓

Point list calculated from the parameters — no typed coordinates

✓

Faces ordered, normals out, closure checked

✓

2 × 2 tiling working at low, middle and high values

✓

One test render: ground plane, one material, lights placed on purpose

A CLOSED MODULE AND A TEST RENDER. THE SHEET FOLLOWS.

BEFORE NEXT CLASS

NEXT

Week 6 starts arrays: 1D, 2D and 3D — a linear array, a nested grid and an XYZ cube array. Your module is the component we repeat.

DUE

P3a is due next class: one sheet, one studio render, settings recorded. DUE WEEK 6 — Sep 29 / 7%.

PRINT

P2b print checkpoint next class. Check geometry, units, orientation and the sliced preview before queueing.

BRING

The editable module file — closed, parameterised, tiling. Next class it becomes the component of every array.